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**Analytical Question:**

Can government highway spending and historical baby-name rankings predict whether a state's most popular baby name will change the following year?

**Comments:**

We are collecting historical baby name ranking and maybe more baby name patterns from official US sites along with other government spending trends. We plan to combine these datasets to analyze their patterns and if we are able to build a prediction model to predict baby name trends in the following year using government spending.

**Data Used:**

SSA State Baby names : https://www.fhwa.dot.gov/policyinformation/statistics/2024/hf2.cfm
Each row represents a state, baby's sex, birth year, first name. Each name is ranked from 1 to 5 in accordance to state. The names are based on social security applications.

Office highway policy information:
https://www.fhwa.dot.gov/policyinformation/statistics/2024/hf2.cfm
Highway funding of 2024 throughout each of the States. Each column has different types of funding cost.

Monthly Consumer Price Index
https://www.bls.gov/cpi/tables/supplemental-files/

This contains the price index for many common items and categories that people purchase for each month from December 2024 through December 2025. It is based on the spending for all urban consumers. The Relative Importance represents the weight applied to it when calculating overall price while the indexes represent how much it costs relative to 2982-84 as a percentage.

Historical CPI-U, December 2025 (XLSX)

This data is just the overall index for each month in the US from 1913 to 2025.

Monthly Treasury Statement (MTS)

https://fiscaldata.treasury.gov/datasets/monthly-treasury-statement/summary-of-receipts-outlays-and-the-deficit-surplus-of-the-u-s-government

This data set contains how much money the US government receives and spends, and whether or not it is a surplus or a deficit. The data is organized by month across different years. We are deciding to focus on recent years around 2024-2026

**Proposed Solution:**

We plan to use the following models to help with our predictions

1. Baseline Model: predict wether the most popular name will changed based on previous years naming pattern
2. Classifier 1 : Logistic Regression to predict patterns
3. Classifier 2: Knn finding the nearest K neighbor to predict patterns

**Proposed evaluation:**

We plan to use folds, divide our data into training and testing sets. We will also use cross-validation to compare the model and may use GridSearchCV to find the best value of K for KNN

Since this problem is similar to a classification problem, we can evaluate the models using accuracy , precision, recall, F1-score, confusion matrix.

**Focus:**

I plan to look into animation and data visualization. I want to be able to display our results in a meaningful way that would represent the relationship well and for information to come across well to the audience.